

Homework #0

Probability and Random Variables

Introduction to Information Theory 2026 · EC5203 / EC4310 / MM4450 · Prof. Heung-No Lee ·
GIST

How to submit. Put your solution in the `submission/` folder of your assignment repository: a single PDF (`submission/hw0.pdf` — scanned handwriting is fine if legible), or Markdown with L^AT_EX (`submission/hw0.md`). Commit and push before the deadline (23:59 KST); the last commit before the deadline is graded. If you worked as a group (max 3), list all members at the top — each member still submits in their own repository.

Questions. Light ones in the AI TA chat on the course homepage; rigorous derivations and proofs on the course GitHub Q&A (mention @claude). Before the deadline you get hints, never solutions.

Problem HW0.1 *Mutually exclusive vs. independent.* The events A and B are mutually exclusive. Can they be independent? Justify your answer from the definitions.

Problem HW0.2 *Probability, random variable, distribution.* A coin with $P\{\text{tail}\} = p$ is tossed n times.

- (a) Find the probability of the event that shows k heads in n trials.
- (b) What is the conditional probability that the first toss is head, given that there are 2 heads in n tosses?
- (c) Let X be the random variable denoting the number of heads. Specify the domain and the range of this random variable.
- (d) Sketch the cumulative distribution function of X for $n = 6$ and $p = 0.1$.

Problem HW0.3 *Balls into pockets.* A box has 10 pockets arranged in 2 rows and 5 columns. Two balls are thrown into the box in sequence; a ball lands in any pocket with equal probability, no pocket can hold two balls, and no ball lands outside.

- (a) What is the probability that both balls land in the same column?
- (b) What is the probability that both balls land in the same row?
- (c) What is the probability that the two balls are separated into both different row and different column?
- (d) Is there any other case? Justify your answer.

Problem HW0.4 *Joint distribution and conditional probability.* Two i.i.d. binary random variables $X_1, X_2 \in \{1, -1\}$ take value 1 with probability p and -1 with probability $1 - p$.

- (a) What is the conditional probability $P(X_1 = 1 \mid X_2 = 1)$?
- (b) Now consider a series of binary random variables $X_1, X_2, X_3, \dots$: X_1 produces equally likely outcomes; the second and the rest are i.i.d. producing 1 with probability p and -1 with probability $1 - p$, where p is determined at the first experiment: $p = 1/2$ if $X_1 = 1$, and $p = 1/4$ if $X_1 = -1$. What is $P\{X_4 = 1\}$? What about $P\{X_1 + X_2 = 2\}$?

Problem HW0.5 *Binary observations through noise.* Let θ, e_1, e_2 be mutually independent binary $\{0, 1\}$ random variables with $P_\theta = P(\theta=1)$, $P_{e_1} = p_1$, $P_{e_2} = p_2$, and define

$$U = \theta + e_1 \pmod{2}, \quad V = \theta + e_2 \pmod{2}.$$

- (a) For $P_\theta = 0.6$, $p_1 = 0.1$, $p_2 = 0.2$, find the joint distribution $P(U=u, V=v)$.
- (b) Repeat with $P_\theta = 0.9$, $p_1 = 0.01$, $p_2 = 0.02$.

Problem HW0.6 *Urn problem.* A box contains m white balls and n black balls. Balls are drawn at random one at a time without replacement. Find the probability of encountering a white ball by the k -th draw.

Problem HW0.7 *Total probability and Bayes' theorem.* A test for prostate cancer is known to be 95% accurate. A person took the test and the result came out positive. Suppose the person comes from a population of one million, in which 20,000 people suffer from the disease. What can we conclude about the probability that the person under test has that particular cancer?